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United States Patent	12418875
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Song; Injae et al.

Electronic device for managing PDU session and operating method thereof

Abstract

An electronic device is provided. The electronic device includes at least one processor, which may be configured to establish at least one first protocol data unit (PDU) session in a state of being registered in a 5th generation system (5GS), store first information related to the at least one first PDU session, perform a procedure for registering with an evolved packet system (EPS), based on at least one first trigger causing a system fallback from the 5GS to the EPS, perform a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, compare second information related to a second PDU session, with the first information, perform an operation of establishing, a third PDU session when the third PDU session not comprised in the second PDU session is identified, and perform an operation of modifying, a fourth PDU session.

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Appl. No.: 17/667107

Filed: February 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220272652 A1	Aug. 25, 2022

Foreign Application Priority Data

KR	10-2021-0021895	Feb. 18, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/KR2022/000522 20220112 PENDING child-doc US 17667107

Publication Classification

Int. Cl.: H04W76/10 (20180101); H04W60/04 (20090101); H04W92/02 (20090101)

U.S. Cl.:

CPC H04W60/04 (20130101); H04W76/10 (20180201); H04W92/02 (20130101)

Field of Classification Search

CPC: H04W (60/04); H04W (76/10); H04W (92/02); H04W (88/06)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365(c), of an International application No.

PCT/KR2022/000522, filed on Jan. 12, 2022, which is based on and claims the benefit of a Korean patent application number 10-2021-0021895, filed on Feb. 18, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(1) The disclosure relates to an electronic device for managing a protocol data unit (PDU) session and an operating method thereof. More particularly, the disclosure relates to an electronic device for performing PDU session establishment and/or modification and an operating method thereof.

2. Description of Related Art

(2) To support various application scenarios based on an Internet protocol (IP) multimedia subsystem (IMS) voice service, a new radio (NR) connected to a user equipment (UE) and a 5th generation core (5GC) network may support a radio access technology (RAT) fallback or an evolved packet system (EPS) fallback (evolved-universal terrestrial radio access network (E-UTRAN) connected to EPC system fallback). For example, the UE connected to the NR connected to the 5GC may be handed over (or directed or redirected) to an evolved universal mobile telecommunications system (UMTS) terrestrial radio access (E-UTRA) connected to the 5GC, which may be referred to as a RAT fallback. Alternatively, the UE connected to the NR connected to the 5GC may be handed over or redirected to an EPS (e.g., an E-UTRA connected to an evolved packet core (EPC)), which may be referred to as a system fallback.

(3) A serving access and mobility function (AMF) for the UE may notify the UE that an IMS voice through a PS session is supported during a registration procedure of the UE. When a request for establishment of a quality of service (QoS) flow for an IMS voice reaches a next-generation radio access network (NG-RAN), the NG-RAN responds to report the rejection of the request for the

establishment and may trigger one of the following procedures, based on UE capability, N26 possibility, a network configuration, and a radio condition. Redirection to EPS Handover to EPS Redirection to E-UTRA connected to 5GC Handover to E-UTRA connected to 5GC

(4) Based on the above description, the UE may access the EPS and/or E-UTRA to perform an IMS service. Alternatively, the UE may perform a fallback in order to perform an emergency service in addition to the IMS service. After the fallback, the UE may return to the NR connected to the 5GC.

(5) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

(6) As described above, when a service (e.g., an Internet protocol (IP) multimedia subsystem (IMS) voice service or an emergency service) in a fallback system is completed after a fallback is performed, a user equipment (UE) may return to new radio (NR) connected to a 5th generation core (5GC) network. The UE may perform the system fallback in a state of establishing at least one protocol data unit (PDU) session based on 5th generation (5G) communication before the fallback. In this case, at least some PDU sessions may be converted into a packet data network (PDN) connection. In addition, when the UE returns to a 5G system (5GS) from a fallback evolved packet system (EPS), at least some PDN connections may be converted into a PDU session. However, at least some of the PDU sessions before the fallback are likely to be lost, or at least some of the PDU sessions are likely to have a change in characteristic. Accordingly, when performing the fallback and then returning to the 5GS, the UE is likely to be unable to use at least some of the PDU sessions established before the fallback.

(7) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device and an operating method capable of storing information related to a PDU session before a fallback, comparing the PDU session with a PDU session identified after returning to a 5th generation system (5GS), and establishing and/or modifying the PDU session, based on a comparison result.

(8) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(9) In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes at least one processor, wherein the at least one processor may be configured to establish at least one first PDU session in a state of being registered in a 5GS, store first information related to the at least one first PDU session, perform a procedure for registering with an EPS, based on at least one first trigger causing a system fallback from the 5GS to the EPS, perform a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, compare second information related to at least one second PDU session, established in association with registration with the 5GS, with the first information, perform an operation of establishing, based on a result of comparing the first information with the second information, at least one third PDU session when the at least one third PDU session not included in the at least one second PDU session among the at least one first PDU session is identified, and perform an operation of modifying, based on the result of comparing the first information and the second information, at least one fourth PDU session when the at least one fourth PDU session having a same identification information as the at least one second PDU session among the at least one first PDU session and having a characteristic different from a characteristic of the at least one second PDU session among the at least one first PDU session is identified.

(10) In accordance with another aspect of the disclosure, an operating method of an electronic device is provided. The operating method includes establishing at least one first PDU session in a state of being registered in a 5GS, storing first information related to the at least one first PDU session, performing a procedure for registering with an EPS, based on at least one first trigger

causing a system fallback from the 5GS to the EPS, performing a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, comparing second information related to at least one second PDU session, established in association with registration with the 5GS, with the first information, performing an operation of establishing, based on a result of comparing the first information with the second information, at least one third PDU session when the at least one third PDU session not included in the at least one second PDU session among the at least one first PDU session is identified, and performing an operation of modifying, based on the result of comparing the first information and the second information, at least one fourth PDU session when the at least one fourth PDU session having a same identification information as the at least one second PDU session and having a characteristic different from a characteristic of the at least one second PDU session among the at least one first PDU session is identified.

(11) According to various embodiments, it is possible to provide an electronic device and an operating method thereof which are capable of storing information related to a PDU session before a fallback, comparing the PDU session with a PDU session identified after returning to a 5GS, and establishing and/or modifying the PDU session, based on a comparison result. Accordingly, the electronic device may equally use at least one PDU session, which is used in the 5GS before the fallback, after the fallback and then the return.

(12) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

(3) FIGS. 2A and 2B are block diagrams illustrating an electronic device for supporting communication via a legacy network and communication via a 5th generation (5G) network according to various embodiments of the disclosure;

(4) FIG. 3 illustrates the structure of a wireless communication system supporting interworking according to an embodiment of the disclosure;

(5) FIGS. 4A and 4B illustrate a protocol data unit (PDU) session and a packet data network (PDN) connection established in an electronic device according to various embodiments of the disclosure;

(6) FIG. 5 is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure;

(7) FIGS. 6A and 6B illustrate a PDU session and a PDN connection established in an electronic device according to various embodiments of the disclosure;

(8) FIGS. 7A and 7B are flowcharts illustrating a PDU session establishment process according to various embodiments of the disclosure;

(9) FIG. 7C is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure;

(10) FIG. 8A is a flowchart illustrating an evolved packet system (EPS) fallback according to an embodiment of the disclosure;

(11) FIG. 8B is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure;

(12) FIG. 9 is a flowchart illustrating a handover from an EPS using an N26 interface to a 5GS

according to an embodiment of the disclosure;

(13) FIG. **10** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure;

(14) FIG. **11** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure; and

(15) FIG. **12** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure.

(16) Throughout the drawings, like reference numerals will be understood to refer to like parts, components, and structures.

DETAILED DESCRIPTION

(17) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

(18) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(19) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

(20) FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

(21) Referring to FIG. **1**, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(22) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing

unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(23) The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(24) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(25) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(26) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(27) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(28) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(29) The audio module **170** may convert a sound into an electrical signal and vice versa. According

to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

(30) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(31) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(32) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

(33) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(34) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(35) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(36) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(37) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g.,

LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(38) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(39) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(40) According to various embodiments, the antenna module **197** may form a mm Wave antenna module. According to an embodiment, the mm Wave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(41) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(42) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should

perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(43) FIGS. 2A and 2B are block diagrams illustrating an electronic device for supporting communication via a legacy network and communication via a 5G network according to various embodiments of the disclosure.

(44) Referring to FIG. 2A depicting block diagram **200**, an electronic device **101** may include a first communication processor **212**, a second communication processor **214**, a first radio-frequency integrated circuit (RFIC) **222**, a second RFIC **224**, a third RFIC **226**, a fourth RFIC **228**, a first radio-frequency front end (RFFE) **232**, a second RFFE **234**, a first antenna module **242**, a second antenna module **244**, a third antenna module **246**, and antennas **248**. The electronic device **101** may further include a processor **120** and a memory **130**. A second network **199** may include a first cellular network **292** and a second cellular network **294**. According to another embodiment, the electronic device **101** may further include at least one of the components illustrated in FIG. 1, and the second network **199** may further include at least one different network. According to an embodiment, the first communication processor **212**, the second communication processor **214**, the first RFIC **222**, the second RFIC **224**, the fourth RFIC **228**, the first RFFE **232**, and the second RFFE **234** may form at least part of a wireless communication module **192**. According to another embodiment, the fourth RFIC **228** may be omitted or may be included as part of the third RFIC **226**.

(45) The first communication processor **212** may establish a communication channel in a band to be used for wireless communication with the first cellular network **292** and may support communication via a legacy network through the established communication channel. According to various embodiments, the first cellular network may be a legacy network including a second-generation (2G), third-generation (3G), fourth generation (4G), or long-term evolution (LTE) network. The second communication processor **214** may establish a communication channel corresponding to a specified band (e.g., about 6 GHz to about 60 GHz) in a band to be used for wireless communication with the second cellular network **294** and may support communication via a 5G network through the established communication channel. According to various embodiments, the second cellular network **294** may be a 5G network defined by the 3GPP. In addition, according to an embodiment, the first communication processor **212** or the second communication processor **214** may establish a communication channel corresponding to a different specified band (e.g., about 6 GHz or lower) in the band to be used for wireless communication with the second cellular network **294** and may support communication via a 5G network through the established communication channel.

(46) The first communication processor **212** may transmit and receive data to and from the second communication processor **214**. For example, data classified to be transmitted through the second

cellular network **294** may be changed to be transmitted through the first cellular network **292**. In this case, the first communication processor **212** may receive the data to be transmitted from the second communication processor **214**. For example, the first communication processor **212** may transmit and receive data to and from the second communication processor **214** through an inter-processor interface **213**. The inter-processor interface **213** may be configured, for example, as a universal asynchronous receiver/transmitter (UART, e.g., high-speed UART (HS-UART)) or peripheral component interconnect express (PCIe) bus interface but is not limited in type. Alternatively, the first communication processor **212** and the second communication processor **214** may exchange control information and packet data information, for example, using a shared memory. The first communication processor **212** may transmit and receive various types of information, such as sensing information, output strength information, and resource block (RB) allocation information, to and from the second communication processor **214**.

(47) The first communication processor **212** may not be directly connected to the second communication processor **214** depending on a configuration. In this case, the first communication processor **212** may transmit and receive data to and from the second communication processor **214** via a processor **120** (e.g., an application processor). For example, the first communication processor **212** and the second communication processor **214** may transmit and receive data to and from each other through the processor **120** (e.g., the application processor) and an HS-UART interface or a PCIe interface, but the type of an interface is not limited. Alternatively, the first communication processor **212** and the second communication processor **214** may exchange control information and packet data information using the processor **120** (e.g., the application processor) and a shared memory.

(48) According to an embodiment, the first communication processor **212** and the second communication processor **214** may be configured in a single chip or a single package. According to various embodiments, the first communication processor **212** or the second communication processor **214** may be configured along with the processor **120**, an auxiliary processor **123**, or a communication module **190** in a single chip or a single package. For example, referring to FIG. **2B** depicting a block diagram **200**, an integrated communication processor **260** may support functions for communication with both the first cellular network **292** and the second cellular network **294**.

(49) In transmission, the first RFIC **222** may convert a baseband signal generated by the first communication processor **212** into a radio-frequency (RF) signal of about 700 MHz to about 3 GHz used for the first cellular network **292** (e.g., a legacy network). In reception, an RF signal may be obtained from the first cellular network **292** (e.g., the legacy network) through an antenna (e.g., the first antenna module **242**) and may be preprocessed by an RFFE (e.g., the first RFFE **232**). The first RFIC **222** may convert the preprocessed RF signal into a baseband signal to be processed by the first communication processor **212**.

(50) In transmission, the second RFIC **224** may convert a baseband signal generated by the first communication processor **212** or the second communication processor **214** into an RF signal (hereinafter, “5G sub-6 RF signal”) in a sub-6 band (e.g., about 6 GHz or lower) used for the second cellular network **294** (e.g., a 5G network). In reception, a 5G sub-6 RF signal may be obtained from the second cellular network **294** (e.g., the 5G network) through an antenna (e.g., the second antenna module **244**) and may be preprocessed by an RFFE (e.g., the second RFFE **234**). The second RFIC **224** may convert the preprocessed 5G sub-6 RF signal into a baseband signal to be processed by a corresponding communication processor of the first communication processor **212** or the second communication processor **214**.

(51) The third RFIC **226** may convert a baseband signal generated by the second communication processor **214** into an RF signal (hereinafter, “5G above-6 RF signal”) in a 5G above-6 band (e.g., about 6 GHz to about 60 GHz) used for the second cellular network **294** (e.g., the 5G network). In reception, a 5G above-6 RF signal may be obtained from the second cellular network **294** (e.g., the 5G network) through an antenna (e.g., an antenna **248**) and may be preprocessed by the third RFFE

236. The third RFIC **226** may convert the preprocessed 5G above-6 RF signal into a baseband signal to be processed by the second communication processor **214**. According to an embodiment, the third RFFE **236** may be configured as part of the third RFIC **226**.

(52) According to an embodiment, the electronic device **101** may include a fourth RFIC **228** separately from the third RFIC **226** or as at least part of the third RFIC **226**. In this case, the fourth RFIC **228** may convert a baseband signal generated by the second communication processor **214** into an RF signal (hereinafter, "IF signal") in an intermediate frequency band (e.g., about 9 GHz to about 11 GHz) and may transmit the IF signal to the third RFIC **226**. The third RFIC **226** may convert the IF signal into a 5G above-6 RF signal. In reception, a 5G above-6 RF signal may be received from the second cellular network **294** (e.g., the 5G network) through an antenna (e.g., the antenna **248**) and may be converted into an IF signal by the third RFIC **226**. The fourth RFIC **228** may convert the IF signal into a baseband signal to be processed by the second communication processor **214**.

(53) According to an embodiment, the first RFIC **222** and the second RFIC **224** may be configured as at least part of a single chip or a single package. According to various embodiments, in FIG. **2A** or FIG. **2B**, when the first RFIC **222** and the second RFIC **224** are configured as a single chip or a single package, the first RFIC **222** and the second RFIC **224** may be configured as an integrated RFIC. In this case, the integrated RFIC may be connected to the first RFFE **232** and the second RFFE **234**, thus converting a baseband signal into a signal in a band supported by the first RFFE **232** and/or the second RFFE **234**, and transmitting the converted signal to one of the first RFFE **232** and the second RFFE **234**. According to an embodiment, the first RFFE **232** and the second RFFE **234** may be configured as at least part of a single chip or a single package. According to an embodiment, at least one of the first antenna module **242** or the second antenna module **244** may be omitted or may be combined with a different antenna module to process corresponding RF signals in a plurality of bands.

(54) According to an embodiment, the third RFIC **226** and the antenna **248** may be disposed on the same substrate to form a third antenna module **246**. For example, the wireless communication module **192** or the processor **120** may be disposed on a first substrate (e.g., a main PCB). In this case, the third RFIC **226** may be disposed in a portion (e.g., a lower surface) of a second substrate (e.g., a sub-PCB) separate from the first substrate, and the antenna **248** may be disposed in another portion (e.g., an upper surface), thereby forming the third antenna module **246**. The third RFIC **226** and the antenna **248** may be disposed on the same substrate, thereby reducing the length of a transmission line therebetween, which may reduce loss (e.g., attenuation) of, for example, a signal in a high-frequency band (e.g., about 6 GHz to about 60 GHz) used for communication via a 5G network due to the transmission line. Accordingly, the electronic device **101** may improve the quality or speed of communication with the second cellular network **294** (e.g., the 5G network).

(55) According to an embodiment, the antenna **248** may be configured as an antenna array including a plurality of antenna elements which can be used for beamforming. In this case, the third RFIC **226** may include, for example, a plurality of phase shifters **238** corresponding to the plurality of antenna elements as part of the third RFFE **236**. In transmission, each of the plurality of phase shifters **238** may convert the phase of a 5G above-6 RF signal to be transmitted to an external device (e.g., a base station of the 5G network) of the electronic device **101** through a corresponding antenna element. In reception, each of the plurality of phase shifters **238** may convert the phase of a 5G above-6 RF signal received from the outside through a corresponding antenna element into the same phase or substantially the same phase. This enables transmission or reception through beamforming between the electronic device **101** and the outside.

(56) The second cellular network **294** (e.g., the 5G network) may operate independently of the first cellular network **292** (e.g., the legacy network) (e.g., stand-alone (SA) mode) or may be connected to operate (e.g., non-stand-alone (NSA) mode). For example, the 5G network may have only an access network (e.g., a 5G radio access network (RAN) or a next-generation RAN (NG RAN)) and

may not have a core network (e.g., a next-generation core (NGC)). In this case, after accessing the access network of the 5G network, the electronic device **101** may access an external network (e.g., the Internet) under control of a core network (e.g., an evolved packet core (EPC)) of the legacy network. Protocol information (e.g., LTE protocol information) for communication with the legacy network or protocol information (e.g., new radio (NR) protocol information) for communication with the 5G network may be stored in a memory **230** and may be accessed by a different component (e.g., the processor **120**, the first communication processor **212**, or the second communication processor **214**).

(57) FIG. **3** illustrates the structure of a wireless communication system supporting interworking between networks according to an embodiment of the disclosure.

(58) According to various embodiments of the disclosure, a standard specification (e.g., Technical Specification (TS) 23.501, TS 23.502, and TS 23.503) defined by the International Telecommunication Union (ITU) or the 3rd Generation Partnership Project (3GPP) may be referred to for a network technology. Each component included in the network structure of FIG. **3** may refer to a physical entity or may refer to software for performing an individual function or hardware combined with the software. Referring to FIG. **3**, reference symbols expressed as Nx, such as N1, N2, N3, and the like, indicate known interfaces between network functions (NFs) in a 5G core network (CN), and a standard specification (TS 23.501) may be referred to for a relevant description, thus obviating the need for a detailed description. In the following description of FIG. **3**, NFs not directly related to various embodiments of the disclosure will not be illustrated/described.

(59) FIG. **3** illustrates an example of the structure of interworking between a 5GS and an EPS in a non-roaming situation. The 5GS may include a new radio (NR) base station (NG-radio access node (RAN) or next-generation node B (gNB)) **303** for radio access of a UE **301** (e.g., the electronic device **101**), an access and mobility management function (AMF) **305**, a session management function (SMF), a user plane function (UPF), a policy control function (PCF), a network slicing selection function (NSSF), a unified data management (UDM), a unified data repository (UDR), and the like. The EPS may include an E-UTRA base station (evolved universal mobile telecommunications system (UMTS) terrestrial radio access network (E-UTRAN) **319** or evolved node B (eNB)), a mobility management entity (MME) **317**, a serving gateway (SGW) **315**, a packet data network gateway (PGW, including a PGW-U and a PGW-C), a policy and charging rule function (PCRF), and a home subscriber server (HSS).

(60) According to an embodiment, the AMF **305** and the MME **317** are network functions (NFs) that manage radio network access and mobility for the UE. The SMF, the SGW, and the PGW are NFs that manage a session for the UE, and session information includes quality-of-service (QoS) information, charging information, and information for packet processing. The UPF and the PGW are NFs that process user plane traffic (e.g., User Plane traffic), and are controlled by the SMF and the SGW. The PCF and the PCRF are NFs that manage an operator policy and/or a public land mobile network (PLMN) policy for providing a service in a wireless communication system. In addition, the PCF may be divided into a PCF in charge of an access and mobility (AM) policy and a UE policy and a PCF in charge of a session management (SM) policy. The PCF in charge of the AM/UE policy and the PCF in charge of the SM policy may be logically or physically separate NFs or may be logically or physically one NF. The UDM and the HSS are NFs that store and manage subscriber information (UE subscription) about the UE. The UDR is an NF or database (DB) that stores and manages data. The UDR may store subscription information about the UE and may provide the subscription information about the UE to the UDM. Further, the UDR may store operator policy information and may provide the operator policy information to the PCF. The NSSF may be an NF that performs a function of selecting a network slice instance serving the UE or determining network slice selection assistance information (NSSAI).

(61) An instance may refer to a state in which an NF exists as a code of software and can

implement a function of the NF by being allocated a physical or/and logical resource from a computing system in order to perform the function of the NF in the physical computing system (e.g., a specific computing system existing in a core network). For example, an AMF instance, an SMF instance, and an NSSF instance may refer to states in which physical and/or logical resources for AMF, SMF, and NSSF operations are respectively allocated from the specific computing system existing in the core network and can be used. Therefore, in a case where physical AMF, SMF, and NSSF devices exist and in an AMF instance, an SMF instance, and an NSSF instance in which physical and/or logical resources for AMF, SMF, and NSSF operations are allocated from a specific computing system existing in a network and are used, the same operation may be performed.

(62) The UDM of the 5GS and the HSS of the EPS may be configured as one combo node (referred to as UDM+HSS node) **307**. The UDM+HSS node **307** may store the subscriber information about the UE. The SMF of the 5GS and the PGW-C of the EPS may be configured as one combo node (referred to as SMF+PGW-C node) **311**. The PCF of the 5GS and the PCRF of the EPS may be configured as one combo node (referred to as PCF+PCRF) **309**. The UPF of the 5GS and the PGW-U of the EPS may be configured as one combo node (referred to as UPF+PGW-C) **313**. The UE **301** may access the MME **317** of the EPS through the E-UTRAN **319** (e.g., a base station) and may use an EPS network service. In addition, the UE may access the AMF **305** of the 5GS through the NR base station and may use a 5GS network service.

(63) As described, one NF or network entity may support different network systems at the same time, and this NF, network node or network entity may be referred to as the foregoing combo node, a combo NF, a combined node, a combined NF, an interworking node, or an interworking NF. A function of an NF illustrated as a combo node may be implemented through interworking between two or more network entities. In addition, for convenience of illustration and description, an NF simultaneously supporting different network systems may be indicated using a sign “+” or a sign “/”. For example, when the SMF and the PGW-C are configured as one combo node, the combo node may be referred to as a PGW-C/SMF, a PGW-C+SMF, an SMF/PGW-C, or an SMF+PGW-C.

(64) The UE establishes a session by accessing a data network (e.g., a network providing an Internet service) through the 5GS or the EPS and may distinguish each data network using an identifier, such as a data network name (DNN) or an access point name (APN). To distinguish a data network, the 5GS may use a data network name (DNN), and the EPS may use an access point name (APN). The DNN and the APN may be used to determine at least one of an NF related to a user plane, an interface between NFs, and an operator policy when the UE establishes a session with a network system. The DNN and the APN may be understood as equivalent information and may transmit the same information. The DNN may be used, for example, to select the SMF and the UPF(s) for a PDU session, and may be used to select an interface (e.g., an N6 interface(s)) between the data network and the UPF for the PDU session. In addition, the DNN may be used to determine a policy of a mobile network operator to be applied to the PDU session.

(65) In the 5GS, the UE may be connected to one data network through at least one network slice. When the UE establishes a PDU session for a certain data network in the 5GS and then moves to the EPS, a PDN connection may be established with the data network of the SMF+PGW-C node **311**.

(66) A service-based interface related to the SMF is defined as ‘Nsmf’ in 3GPP TS 23.502, for example, as shown in Table 1.

(67) TABLE-US-00001

Service Name	Service Operations	Semantics	Example
Consumer(s) Nsmf_PDU Session Create Request/Response	V-SMF/I-SMF	Update Request/Response	V-SMF/I-SMF, H-SMF
Release Request/Response	V-SMF/I-SMF	CreateSMContext Request/Response	AMF
UpdateSMContext Request/Response	AMF	ReleaseSMContext Request/Response	AMF
SMContextStatusNotify	Subscribe/Notify	AMF	StatusNotify
Subscribe/Notify	V-SMF/I-SMF	ContextRequest	
Request/Response	AMF, I-SMF, SMF	ContextPush	Request/Response
SMF SendMOData			

(68) In Table 1, 'Nsmf_PDUSession' refers to a service operating in a PDU session, the service includes operations of establishing/deleting/modifying the PDU session, and these operations may be performed through transmission and reception of a PDU session request/response message between the AMF and the SMF. As illustrated in Table 1, in an association creation operation between the AMF and the SMF for supporting a PDU session, the SMF may receive a 'Nsmf_PDUSession_CreateSMContext' request message, which is a PDU session request message, from the AMF and may transmit a 'Nsmf_PDUSession_CreateSMContext' response message to the AMF in response. Related standards may be referred to for other service operations in Table 1, and a detailed description thereof will be omitted.

(69) In the following embodiments, combo nodes, such as a UDM+HSS node, a PCF+PCRF node, a SMF+PGW-C node, and a UPF+PGW-C node, will be described with the term "node" omitted for convenience of description. Further, in the following embodiments, a definition of a message defined in one embodiment may be applied with the same meaning to other embodiments using the same message.

(70) FIGS. 4A and 4B illustrate a PDU session and a PDN connection established in an electronic device according to various embodiments of the disclosure. At least some of operations of the electronic device **101** according to the comparative examples may also be performed by the electronic device **101** according to various embodiments.

(71) The electronic device **101** (e.g., at least one of the processor **120** of FIG. 1, FIG. 2A, or FIG. 2B, the first communication processor **212** of FIG. 2A, the second communication processor **214** of FIG. 2A, or the integrated communication processor **260** of FIG. 2B) may establish at least one PDU session, for example, a first PDU session **401** and a second PDU session **402**, for example, in a state of being registered in a 5GS. For example, the electronic device **101** may transmit a PDU session establishment request to a network. The network may establish a PDU session between the electronic device **101** and the network, based on the PDU session establishment request from the electronic device **101**, and a PDU establishment process will be described with reference to FIG. 7A. Accordingly, in a state in which the first PDU session **401** and the second PDU session **402** are established, the electronic device **101** may perform data transmission and reception, based on the first PDU session **401** and/or the second PDU session **402**. The first PDU session **401** and the second PDU session **402** may be identified by a PDU session ID (e.g., a PDU session identifier (PSI)) and/or a QoS flow identifier (QFI).

(72) Although not shown, each of the first PDU session **401** and/or the second PDU session **402** may include at least one QoS flow and a QoS rule. The electronic device **101** may receive, for example, a QoS flow descriptor from the network, and the QoS flow description may include information related to the QoS flow (e.g., a QoS flow ID (QFI) and a QoS flow parameter list). For example, the QoS flow parameter list may include an EPS bearer identity (EBI). The EBI included in the QoS flow parameter list may be, for example, an identifier for identifying an EPS bearer mapped to the QoS flow, and may be referred to as a mapped EBI. Information related to the EPS bearer mapped to the QoS flow may be referred to as a mapped EPS bearer context.

(73) The electronic device **101** may perform a system fallback to an EPS (hereinafter, referred to as an EPS fallback) in a state being registered to the 5GS. For example, the electronic device **101** may perform an EPS fallback in order to perform a service that causes an EPS fallback (e.g., an IMS voice service or an emergency service). For example, the electronic device **101** may perform an EPS fallback upon receiving a command to hand over to an access network (AN) connected to the EPS from the network or an RRC release message for a redirection to the AN connected to the EPS. An EPS fallback procedure will be described with reference to FIG. 8A. After performing the handover or the redirection, based on the EPS fallback, the electronic device **101** may perform a tracking area update (TAU) procedure or an attach procedure to the EPS. The electronic device **101**

may establish a PDN connection during the procedure. Alternatively, it may be expressed that the electronic device **101** converts the PDU session into a PDN connection, and this case is also expressed as establishing a PDN connection in various embodiments of the disclosure.

(74) For example, referring to FIG. 4A, an electronic device **101** may establish a first PDN connection **411**. The first PDN connection **411** may correspond to the first PDU session **401**. The electronic device **101** may establish an EPS bearer, for example, based on the stored EPS bearer context.

(75) The electronic device **101** establishes the first PDN connection **411** corresponding to the first PDU session **401** as illustrated in FIG. 4A, but may not establish a PDN connection corresponding to the second PDU session **402**. For example, the electronic device **101** may convert the first PDU session **401** into the first PDN connection **411** by changing a characteristic of the first PDU session **401**. For example, the first PDU session **401** may include a DNN of “IMS” and S-NSSAI of “A”, and the second PDU session **402** may include a DNN of “Internet” and the S-NSSAI of “A”. The electronic device **101** may change the DNN of the first PDU session **401** to an APN and may delete the S-NSSAI not supported in LTE, but there is no limitation on examples of a change. Information related to the PDU session used by the electronic device **101** in the 5GS may or may not be supported according to the EPS. For example, 3GPP TS 23.501 suggests that mobility from a 5th generation core (5GC) to an evolved packet core (EPC) does not guarantee that all activated PDU session(s) can be transferred to the EPC. In addition, mobility from the EPC to the 5GC may also not guarantee that all activated PDU session(s) can be transferred to the 5GC. The S-NSSAI of the PDN connection mapped to the PDU session may be configured according to a policy of the network. For example, it may be guaranteed that the existing PDU session is equally configured. According to 3GPP TS 24.501, even though the N26 interface is configured, when there is no mapped session (or connection) even though an N26 interface is configured, local release may occur. For example, it is assumed that the second PDU session **402** includes the DNN of “IMS” and S-NSSAI of “B”. In this case, the electronic device **101** does not establish PDN connections corresponding to the respective PDU sessions **401** and **402**, based on the first PDN connection **411** having the APN of “IMS”, and may transmit and receive data, based on the first PDN connection **411** having the APN of “IMS”. Accordingly, there is a possibility that a PDN connection corresponding to the second PDU session **402** may not be established.

(76) The electronic device **101** may move back to the 5GS after the EPS fallback. For example, the electronic device **101** may receive a command to hand over to the AN connected to the 5GC and may thus perform a procedure for a handover to the AN connected to the 5GC. The electronic device **101** may perform a procedure for registration with the 5GC, and the procedure for the registration will be described with reference to FIG. 9. The electronic device **101** may establish a first PDU session **421**, based on the procedure for the registration with the 5GC or a PDU session establishment procedure after the registration with the 5GC. For example, the electronic device **101** may convert the first PDN connection **411** into the first PDU session **421** by changing a characteristic of the first PDN connection **411**. The first PDU session **421** may have the same characteristic, for example, as the first PDU session **401** previously established. There is a possibility that the electronic device **101** may not establish a PDU session having the same characteristic as the second PDU session **402** previously established after returning to the 5GS. In this case, the electronic device **101** may not be able to continuously use a service used based on the second PDU session **402** after performing the EPS fallback and then returning to the 5GS.

(77) Alternatively, referring to FIG. 4B as another comparative example, an electronic device **101** may perform an EPS fallback in a state in which a first PDU session **401** and a second PDU session **402** are established. The electronic device **101** may establish (or convert) a first PDN connection **411** corresponding to the first PDU session **401** and a second PDN connection **412** corresponding to the second PDU session **402** in a state of being registered in the EPS. In the comparative example of FIG. 4B, compared with a comparative example of FIG. 4A, the second PDN

connection **412** corresponding to the second PDU session **402** may also be established. The electronic device **101** may return to the 5GS. The electronic device **101** may establish a first PDU session **421** corresponding to the first PDN connection **411** and a second PDU session **422** corresponding to the second PDN connection **412**. There is a possibility that a characteristic of the second PDU session **422** may be different from a characteristic of the second PDU session **402** before the EPS fallback. For example, the QFI of a QoS flow included in the second PDU session **402** may be different from the QFI of a QoS flow included in the second PDU session **422**. Since a specific QoS flow before the EPS fallback may be configured in the network to be suitable for a specific service, there is a possibility that the service before the EPS fallback may be affected by a change in the QFI of the QoS flow. In one example, it is assumed that the second PDU session **402** includes a DNN of “Internet” and S-NSSAI of “B”. In this case, the second PDN connection **412** corresponding to the second PDU session **402** may be established to have an APN of “Internet”, but the information of the S-NSSAI may be deleted. Accordingly, after returning to the 5GS, the second PDU session **422** corresponding to the second PDN connection **412** may include the DNN of “Internet” and S-NSSAI of “C” rather than “B”.

(78) Table 2 illustrates a mapping relationship between a PDU session in a 5GS, a PDN connection after an EPS fallback, and a PDU sessions after returning to the 5GS according to various examples.

(79) TABLE-US-00002

5GS PDU1	DNN:IMS	PDN APN: PDU	DNN:IMS	S-NSSAI:A
1' IMS	1" S-NSSAI:A	PDU2 DNN:Internet	PDN APN: PDU	DNN:Internet
S-NSSAI:A	2' Internet	2" S-NSSAI:A	PDU3 DNN:Internet	S-NSSAI:B
PDU4	DNN:Streaming	S-NSSAI:B		

(80) In an example of Table 2, in the initial 5GS, a first PDU session (PDU1) with a DNN of “IMS” and S-NSSAI of “A”, a second PDU session (PDU2) with a DNN of “Internet” and the S-NSSAI of “A”, a third PDU session (PDU3) with the DNN of “IMS” and S-NSSAI of “B”, and a fourth PDU session (PDU4) with a DNN of “streaming” and the S-NSSAI of “B” may be established. After the EPS fallback, the electronic device **101** may establish a first PDN connection (PDN1') with an APN of “IMS” corresponding to the first PDU session (PDU1) and a second PDN connection (PDN2') with an APN of “Internet” corresponding to at least some of the PDU sessions (PDU2, PDU3, and PDU4). After returning to the 5GS, the electronic device **101** may establish a first PDU session (PDU1'') with the DNN of “IMS” corresponding to the first PDN connection (PDN1') and a second PDU connection (PDU2'') with the DNN of “Internet” and the S-NSSAI of “A” corresponding to the second PDN connection (PDN2'). Comparing a pre-EPS fallback and a post-EPS fallback, the third PDU session (PDU3) and the fourth PDU session (PDU4) are lost, and accordingly there is a possibility the electronic device **101** may perform a service based on the third PDU session (PDU3) and the fourth PDU session (PDU4). Alternatively, the electronic device **101** may declare a local release without establishing a PDN connection corresponding to the fourth PDU session (PDU4), and even in this case, there is a possibility that the fourth PDU session PDU4 may be lost.

(81) FIG. 5 is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure. An embodiment of FIG. 5 is described with reference to FIGS. 6A and 6B.

(82) FIGS. 6A and 6B illustrate a PDU session and a PDN connection established in an electronic device according to various embodiments of the disclosure.

(83) Referring to FIG. 5, according to various embodiments, in operation **501**, an electronic device **101** (e.g., at least one of a processor **120** of FIG. 1, FIG. 2A, or FIG. 2B, a first communication processor **212** of FIG. 2A, a second communication processor **214** of FIG. 2A, or an integrated communication processor **260** of FIG. 2B) may establish at least one first PDU session in a state of being registered in a 5GS. For example, the electronic device **101** may transmit a PDU session establishment request to a network, and PDU session establishment will be described with reference to FIG. 7A.

(84) According to various embodiments, in operation **503**, the electronic device **101** may store first information related to the at least one first PDU session. For example, the electronic device **101** may store at least one of network information (e.g., at least one of a PLMN, a TAC, or a cell ID) about the first PDU session, PDU session information (e.g., at least one of a session and service continuity (SSC) mode, an always-on mode, a PDU session type, a DNN, S-NSSAI, a PSI, a QFI, or a mapped EBI), or QoS information (e.g., at least one of a QoS rule or QoS flow information), but the first information is not limited to a specific type. When a plurality of PDU sessions is established, the electronic device **101** may store relevant information about each of the plurality of PDU sessions.

(85) According to various embodiments, in operation **505**, the electronic device **101** may perform a procedure for EPS registration (or EPC registration), based on a first trigger. For example, the electronic device **101** may perform the procedure for EPS registration upon receiving a handover command from the network or an RRC release message including redirection information from the network. For example, the electronic device **101** may detect a cell for an EPS fallback, may perform a random access channel (RACH) procedure, and may perform a TAU procedure or an attach procedure. After performing EPS registration, the electronic device **101** may perform a service requiring an EPS fallback (e.g., an IMS voice service or an emergency service). The electronic device **101** may transmit a TAU request message or an attach request including a mapped EBI during an EPS registration process. The electronic device **101** may receive a TAU accept message from the network, and the TAU accept message may include an EBI. The electronic device **101** may establish a PDN connection between the network and the EBI included in the TAU accept message. The electronic device **101** may store information related to the PDN connection. For example, the electronic device **101** may store, as the information related to the PDN connection, network information (e.g., at least one of a PLMN, a TAC, or a cell ID), PDN connection information (e.g., at least one of an EBI, an APN, or a PDN type), or QoS information (e.g., at least one of a QoS rule in a protocol configuration options (PCO) or a description), but there is no limitation on the information.

(86) According to various embodiments, in operation **507**, the electronic device **101** may perform a procedure for registration with the 5GS, based on a second trigger. For example, after the service requiring the EPS fallback is terminated, the electronic device **101** may receive a handover command in which a cell supporting the 5GS is designated as a target cell from the network. The electronic device **101** may identify reception of the handover command as the second trigger and may perform the procedure for registration with the 5GS. For example, the electronic device **101** may detect the target cell, may perform a RACH procedure, and may perform registration (e.g., a registration procedure) with the 5GS. The procedure for registration with 5GS may include transmission of a registration request message by the electronic device **101**.

(87) According to various embodiments, in operation **509**, the electronic device **101** may compare second information related to at least one second PDU session, established in association with the registration with the 5GS, with the first information. In one example, when an N26 interface is supported in the network, the electronic device **101** may transmit the registration request message including an EBI, for example, the EBI corresponding to the PDN connection established in the EPS registration, to the network. In another example, when the N26 interface is not supported in the network, the electronic device **101** may transmit a PDU session establishment request message including an EBI, for example, the EBI corresponding to the PDN connection established in the EPS registration, to the network after performing the procedure for the registration with the 5GS. Accordingly, the electronic device **101** may establish the at least one second PDU session with the network. The electronic device **101** may compare the second information about the second PDU session with the first information by each item.

(88) According to various embodiments, in operation **511**, the electronic device **101** may determine whether at least one third PDU session not included in the at least one second PDU session among

the at least one first PDU session is identified. When the at least one third PDU session not included in the at least one second PDU session among the at least one first PDU session is identified (Yes in 511), the electronic device **101** may perform an operation of establishing the at least one third PDU session in operation 513. For example, referring to FIG. 6A, the electronic device **101** may establish at least one PDU session, for example, a first PDU session **401** and a second PDU session **402**, in a state being registered with the 5GS. After an EPS fallback, the electronic device **101** may establish a first PDN connection **411** corresponding to the first PDU session **401** but may not establish a PDN connection corresponding to the second PDU session **402**. For the same reason as described with reference to FIG. 4A, the PDN connection corresponding to the second PDU session **402** may not be established. When returning to the 5GS, the electronic device **101** may establish a first PDU session **421** corresponding to the first PDN connection **411**. However, since no PDN connection corresponding to the second PDU session **402** is established, only the first PDU session **421** may be established after returning to the 5GS. The electronic device **101** may compare information related to the PDU sessions before the EPS fallback with information related to the PDU session after the EPS fallback and then the return to the 5GS. The electronic device **101** may identify that the second PDU session **402** among the PDU sessions before the EPS fallback does not exist after the return. The electronic device **101** may establish a second PDU session **423**, based on information related to the second PDU session **402**. For example, the electronic device **101** may transmit a PDU session establishment message to the network, and the PDU session establishment message may include at least part of the information related to the existing second PDU session **402**. Accordingly, after returning to the 5GS, the electronic device **101** may establish both the first PDU session **421** corresponding to the first PDU session **401** and the second PDU session **423** corresponding to the second PDU session **402** and may perform a previously implemented service.

(89) According to various embodiments, in operation 515, the electronic device **101** may identify whether at least one fourth PDU session having the same identification information as the at least one second PDU session among the at least one first PDU session and having a different characteristic therefrom is identified. When the at least one fourth PDU session having the same identification information as the at least one second PDU session among the at least one first PDU session and having the different characteristic therefrom is identified (Yes in 515), the electronic device **101** may perform an operation of modifying the at least one fourth PDU session in operation 517. For example, referring to FIG. 6B, the electronic device **101** may establish at least one PDU session, for example, a first PDU session **401** and a second PDU session **402**, in a state being registered with the 5GS. After an EPS fallback, the electronic device **101** may establish a first PDN connection **411** corresponding to the first PDU session **401** and a second PDN connection **412** corresponding to the second PDU session **402**. When returning to the 5GS, the electronic device **101** may establish a first PDU session **421** corresponding to the first PDN connection **411** and a second PDU session **422** corresponding to the second PDN connection **412**. However, as described with reference to FIG. 4B, a characteristic (e.g., S-NSSAI) of the second PDU session **422** may be different from a characteristic (e.g., S-NSSAI) of the second PDU session **402** before the EPS fallback. As the characteristics of the corresponding PDU sessions (e.g., the second PDU sessions **402** and **422**) before the EPS fallback and after returning to the 5GS are different, the electronic device **101** may perform an operation of modifying the characteristic of the second PDU session **422** established after returning to the 5GS. For example, the electronic device **101** may transmit a PDU session modification request message to the network. The PDU session modification request message may include identification information (e.g., at least one of a PSI, a corresponding EBI, or a QFI) about the PDU session to be modified and modification information but is not limited thereto. Accordingly, a modified second PDU session **423** may be established between the electronic device **101** and the network, and a characteristic of the modified second PDU session **423** may be the same of the characteristic of the second PDU session **402** before the EPS fallback.

Accordingly, the electronic device **101** can perform communication in the same environment as an environment in which the PDU sessions **401** and **402** before the EPS fallback are established, thus performing a service before the EPS fallback.

(90) FIG. 5 shows that operation **511** is performed before operation **515** is performed, which is only one example. According to various embodiments, operation **515** may be performed before operation **511** is performed, or operation **511** and operation **515** may be performed at least simultaneously, and it will be understood by those skilled in the art that there is no limitation on the order of the operations. Further, it will be understood by those skilled in the art that either operation **511** or operation **515** may be omitted in various embodiments.

(91) In various embodiments, the processor **120** may perform operation **511** and/or operation **515** and may request the communication processor (at least one of the first communication processor **212**, the second communication processor **214**, or the integrated communication processor **260**) to perform subsequent operation **513** and/or operation **517**. Alternatively, the communication processor (e.g., at least one of the first communication processor **212**, the second communication processor **214**, or the integrated communication processor **260**) may perform at least one of operation **511**, operation **513**, operation **515**, or operation **517**.

(92) FIGS. 7A and 7B are flowcharts illustrating a PDU session establishment process according to various embodiments of the disclosure.

(93) Referring to FIG. 7A, according to various embodiments, in operation **701**, a UE **700a** (e.g., an electronic device **101**) may transmit a PDU session establishment request message to an AMF **700c** in a state of being registered with a 5GS. For example, a PDU session establishment request is a network access stratum (NAS) message and may include at least one of S-NSSAI, a DNN, a PDU session ID, a requested PDU session type, a requested SSC mode, a 5GSM capability PCO, an SM PDU DN request container, or the number of packet filters. To establish a new PDU session, the UE **700a** may generate a new PDU session ID. The AMF **700c** may receive the PDU session establishment request message, which is the NAS message, through an (R)AN **700b**.

(94) According to various embodiments, in operation **703**, the AMF **700c** may select an SMF **700e** corresponding to the received PDU session establishment request message. For example, the AMF **700c** may select the SMF **700e** corresponding to the S-NSSAI and the DNN of the PDU session establishment request message. When the S-NSSAI is not included in the PDU session establishment request message, the AMF **700c** may select default S-NSSAI. When the S-NSSAI is included in the PDU session establishment request message but the DNN is not included, the AMF **700c** may select a default DNN as a DNN for a requested PDU session.

(95) According to various embodiments, in operation **705**, the AMF **700c** may transmit a Nsmf_PDUSession_CreateSMContext Request to the selected SMF **700e**. For example, the AMF **700c** may transmit the S-NSSAI from allowed NSSAI to the SMF **700e**. In operation **707**, the SMF **700e** may perform subscription retrieval/subscription for updates with a UDM **700g**. For example, the SMF **700e** may perform registration of the UDM **700g** using Nudm_UECM_Registration for a given PDU session. The UDM **700g** may store at least one of a SUPI, an SMF identity, an SMF address and an associated DNN, or a PDU session ID. In operation **709**, the SMF **700e** may transmit an Nsmf_PDUSession_CreateSMContext Response to the AMF **700c**. In operation **711**, a PDU session authentication/authorization procedure may be performed. When a PDU session already exists or when a request is made according to an emergency service, operation **711** may be omitted.

(96) According to various embodiments, when dynamic policy and charging control (PCC) is deployed and a PCF ID is provided by the AMF **700c**, the SMF **700e** may perform PCF selection in operation **713**. In operation **715**, the SMF **700e** may establish a PDU session with a PCF **700f** by performing an SM policy association establishment procedure, and may obtain a default PCC rule for the PDU session or may perform an SM policy association modification procedure. In operation **717**, the SMF **700e** may select the SSC mode for the PDU session and may select a UPF. The SMF

700e may assign an IP address/prefix to the PDU session. In operation **719**, the **SMF 700e** may perform an SM policy association modification procedure and may thus provide a policy control request trigger condition. The **SMF 700e** may perform an N4 session establishment procedure with the selected **UPF 700d**. For example, in operation **731**, the **SMF 700e** may transmit an N4 session establishment/modification request to the **UPF 700d**. In operation **733**, the **UPF 700d** may perform a response by transmitting an N4 session establishment/modification response. Here, CN tunnel information may be provided to the **SMF 700e**. In operation **735**, the **SMF 700e** may transmit **Namf_Communication_NIN2MessageTransfer** to the **AMF 700c**.

Namf_Communication_NIN2MessageTransfer may include information (e.g., a PDU session ID, N2 SM information, and an N1 SM container) related to the PDU session. The N2 SM information may include at least one of the PDU session ID, a QFI(s), a QoS profile(s), CN tunnel information, the S-NSSAI from the allowed NSSAI, a session-AMBR, a PDU session type, user plane security enforcement information, or UE integrity protection maximum data rate. The N1 SM container may include PDU session establishment accept to be transmitted to the **UE 700a**, and the PDU session establishment accept may include least one of a QoS rule(s), a QoS flow level, QoS parameters, QoS rule(s), the SSC mode, the S-NSSAI(s), the DNN, an allocated IPv4 address, an interface identifier, the session-AMBR, a selected PDU session type, a reflective QoS timer, or a P-CSCF address.

(97) Referring to FIG. 7B, according to various embodiments, in operation **737**, an **AMF 700c** may transmit an N2 PDU session request to a (R)AN **700b**, and this request may include the N2 SM information and a NAS message. The NAS message may include the PDU session ID and the N1 SM container, and the N1 SM container may include the PDU session establishment accept. In operation **739**, the (R)AN **700b** may perform AN-specific resource setup. For example, the (R)AN **700b** may perform an AN-specific signaling exchange with the **UE 700a** associated with the information received from the **AMF 700c**. For example, an NG-RAN may perform RRC connection reconfiguration with the **UE 700a**. The (R)AN **700b** may forward a NAS message including the PDU session ID and the N1 SM container (PDU session establishment accept) to the **UE 700a**. In operation **741**, the (R)AN **700b** may transmit an N2 PDU session response to the **AMF 700c**, and this response may include the PDU session ID, Cause and the N2 SM information. The N2 SM information may include the PDU session ID, the AN tunnel information, a list of an accepted/rejected QFI(s), and a user plane enforcement policy notification. Accordingly, first uplink data may be transmitted from the **UE 700a** to the **UPF 700d** in operation **743**.

(98) According to various embodiments, in operation **745**, the **AMF 700c** may transmit an **Nsmf_PDUSession_UpdateSMContext Request** to the **SMF 700e**, and this request may include the N2 SM information and a request type. The **AMF 700c** may forward the N2 SM information, received from the (R)AN **700b**, to the **SMF 700e**. when the N2 SM information includes a list of a rejected QFI, the **SMF 700e** may release a QoS profile associated with the rejected QFI. The **SMF 700e** may perform an N4 session modification procedure with the **UPF 700d**. In operation **747**, the **SMF 700e** may transmit an N4 session modification request to the **UPF 700d**. For example, the **SMF 700e** may provide the AN tunnel information to the **UPF 700d**. In operation **749**, the **UPF 700d** may provide an N4 session modification response to the **SMF 700e**. Subsequently, the **UPF 700d** may transmit a downlink packet directed to the **UE 700a**, which is buffered for the PDU session. Accordingly, in operation **753**, first downlink data may be provided from the **UPF 700d** to the **UE 700a**. In operation **755**, the **SMF 700e** may transmit an **Nsmf_PDUSession_UpdateSMContext Response** to the **AMF 700c**. Accordingly, the **AMF 700c** may forward a relevant event subscribed to by the **SMF 700e**. In operation **757**, the **SMF 700e** may transmit **Nsmf_PDUSession_SMContextStatusNotify** to **AMF 700c**, which may be conditionally performed and may cause release. In operation **759**, the **SMF 700e** may provide an IPV6 address configuration to the **UE 700a** through the **UPF 700d**. When the PDU session type is IPV6 or IPV4v6, the **SMF 700e** may generate an IPV6 router advertisement and may transmit the same to

the UE **700a**. When establishment of the PDU session fails, the SMF **700e** may perform an SM policy modification with the PCF **700f** in operation **761** and may perform unsubscription with the UDM **700g** in operation **763**. The UDM **700g** may update UE context by Nudr_DM_Update. Through the foregoing process, the UE **700a** (e.g., the electronic device **101**) may establish PDU sessions **401** and **402** as shown in FIG. 6A or 6B in a state of being registered with the 5GS, which is described in an embodiment with reference to FIG. 7C.

(99) FIG. 7C is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure.

(100) Referring to FIG. 7C, an electronic device **101** may perform a procedure for registration with a first network **700** corresponding to the 5GS. In operation **771**, the electronic device **101** (e.g., at least one of the processor **120** of FIG. 1, FIG. 2A, or FIG. 2B, the first communication processor **212** of FIG. 2A, the second communication processor **214** of FIG. 2A, or the integrated communication processor **260** of FIG. 2B) may transmit a registration request for registration with the 5GS to the first network **700** corresponding to the 5GS. In operation **773**, the first network **700** may transmit a registration accept to the electronic device **101**. In operation **775**, the electronic device **101** may transmit a registration complete to the first network **700**. The procedure for the registration may be performed, for example, based on 3rd generation partnership project (3GPP) technical specification (TS) **23.501**.

(101) According to various embodiments, the electronic device **101** may perform a procedure **780** for establishing a first PDU session of a DNN of “Internet” in a state of being registered with the 5GS. In operation **781**, the electronic device **101** may transmit a PDU session establishment request message to the first network **700**. For example, the PDU session establishment request message may include a PDU session ID (e.g., PSI=5) and information about the DNN of “Internet”. For example, the PDU session establishment request message may include S-NSSAI. In the first network **700**, for example, operations as described with reference to FIGS. 7A and 7B may be performed. In operation **783**, the first network **700** may transmit a PDU session establishment accept message to the electronic device **101**. For example, the PDU session establishment accept message may include the PDU session ID and a mapped EBI (e.g., EBI=5).

(102) According to various embodiments, the electronic device **101** may perform a procedure **790** for establishing a second PDU session using a UE route selection policy (URSP) rule in the state of being registered with the 5GS. In operation **791**, the electronic device **101** may transmit a PDU session establishment request message to the first network **700**. For example, the PDU session establishment request message may include a PDU session ID (e.g., PSI=6), S-NSSAI identified based on the URSP rule, and information about a DNN corresponding to the S-NSSAI. In the first network **700**, for example, operations as described with reference to FIGS. 7A and 7B may be performed. In operation **793**, the first network **700** may transmit a PDU session establishment accept message to the electronic device **101**. For example, the PDU session establishment accept message may include the PDU session ID and a QoS flow description, and the QoS flow description may include a mapped EBI (e.g., EBI=6). In operation **795**, the electronic device **101** may store information related to the PDU sessions, for example, information related to the first PDU session and information related to the second PDU session. As described above, the electronic device **101** may establish the PDU sessions in the state of being registered with 5GS and may perform an EPS fallback. Hereinafter, a process for an EPS fallback according to an embodiment will be described with reference to FIGS. 8A and 8B.

(103) FIG. 8A is a flowchart illustrating an EPS fallback according to an embodiment of the disclosure.

(104) Referring to FIG. 8A, according to various embodiments, a UE **800a** (e.g., an electronic device **101**) may camp on an NG-RAN **800b** in a 5GS. In operation **831**, an MO or MT IMS voice session in a 5GS may be established by the UE **800a**, a network (e.g., **800b**, **800c**, **800d**, **800e**, **800f**, **800g**, and **800h**), and an IMS server **800i**, and establishment of a QoS flow for a voice may

be initiated. In operation **833**, the second network (e.g., **800b**, **800c**, **800d**, **800e**, **800f**, **800g**, and **800h**) may perform PDU session modification for setting up the QoS flow for an IMS voice. An NG-RAN **800b** may support an EPS fallback for the IMS voice, and in operation **835**, the NG-RAN **800b** may determine to trigger the fallback. The NG-RAN **800b** may determine to trigger the fallback, based on UE capability, an indication of “Redirection for EPS fallback for voice is possible” from an AMF **800d**, a network configuration (e.g., a setting for N26 availability), and a radio condition. The NG-RAN **800b** may optionally start a measurement report request from the UE **800a** including an E-UTRAN as a target. In operation **837**, the NG-RAN **800b** may reject the PDU session modification for setting up the QoS flow for the IMS voice and may indicate that a movement by the fallback for the IMS voice is in progress.

(105) According to various embodiments, in operation **839**, the NG-RAN **800b** may initiate a handover or an inter-system redirection to an EPS in consideration of the UE capability. The UE **800a** may receive a handover command or an RRC release message including redirection information from the network. The UE **800a** may perform a handover procedure or a redirection procedure. For example, the UE **800a** may retrieve a target cell, based on information included in the handover command, and may perform an RACH procedure with the target cell, based on a retrieval result. For example, the UE **800a** may retrieve a cell, based on the information included in the RRC release message, and may perform an RACH procedure with the selected cell, based on a retrieval result. When the UE **800a** performs a handover or performs a redirection procedure in which an N26 interface is supported, the UE **800a** may perform a tracking area update (TAU) procedure in operation **841**. When the UE **800a** performs a redirection procedure in which the N26 interface is not supported, the UE **800a** performs an attach procedure using a PDN connectivity request of a request type of “handover” in operation **843**. When the movement to the EPS is completed, the network may perform a PDN connection modification procedure for setting up a dedicated bearer for the voice in operation **845**.

(106) FIG. **8B** is a flowchart illustrating an EPS fallback according to an embodiment of the disclosure.

(107) According to various embodiments, it is assumed that an electronic device **101** (e.g., at least one of the processor **120** of FIG. **1**, FIG. **2A**, or FIG. **2B**, the first communication processor **212** of FIG. **2A**, the second communication processor **214** of FIG. **2A**, or the integrated communication processor **260** of FIG. **2B**) has established two PDU sessions in a state of being registered in a 5GS as in FIG. **7C** before operation **801** of FIG. **8B** is performed. Referring to FIG. **8B**, in operation **801**, the electronic device **101** may receive an RRC release message for an EPS fallback from a first network **700** corresponding to the 5GS. Although FIG. **8B** shows that the electronic device **101** performs a redirection procedure, it will be understood by those skilled in the art that the electronic device **101** may perform a handover procedure according to various embodiments. The electronic device **101** may perform cell retrieval and cell selection upon receiving the RRC release message, and may perform a RACH procedure with a selected cell. In operation **803**, the electronic device **101** may transmit a tracking area update (TAU) request through a second network **800** corresponding to an EPS, for example, a cell with which an RRC connection is established. The TAU request may include, for example, an EBI (e.g., EBI=5) of a PDN connection. In operation **805**, the electronic device **101** may receive a tracking area update (TAU) accept from the second network **800**. The TAU accept may include, for example, the EBI (e.g., EBI=5) corresponding to the PDN connection. For example, in FIG. **8B**, it is assumed that PDN connections corresponding to some PDU sessions among a plurality of PDU sessions are not established. For example, among a first PDU session having EBI=5 and PSI=5 and a second PDU session having EBI=6 and PSI=6, only a PDN connection corresponding to the first PDU session is established, while a PDN connection corresponding to the second PDU session may not be established. The PDN connection may have the EBI of 5. In operation **807**, the electronic device **101** may perform an IMS voice-based call, based on the PDN connection. In operation **809**, the electronic device **101** may

terminate the call. As described above, the electronic device **101** may perform the EPS fallback, may perform the IMS voice-based call, and may return to the 5GS when the call is terminated. Hereinafter, a process of returning to the 5GS according to an embodiment will be described with reference to FIGS. **9** and **10**.

(108) FIG. **9** is a flowchart illustrating a handover from an EPS using an N26 interface to a 5GS according to an embodiment of the disclosure. The N26 interface may provide seamless session continuity in a single registration mode. For example, in home-routed roaming, an SMF+PGW-C **900i** may receive a PDU session ID from a UE **900a** and may provide a different 5G QOS parameter to the UE **900a**, which may be applied even when a home PLMN (HPLMN) performs an interworking procedure without the N26 interface. In an embodiment of FIG. **9**, it is assumed that the electronic device **101** is in a state of being registered in the EPS after an EPS fallback.

(109) Referring to FIG. **9**, according to various embodiments, in operation **901**, a UE **900a** (e.g., an electronic device **101**) may transmit and receive uplink and downlink user plane PDUs (UL and DL UP PDUs) to and from a UPF+PGW-U **900j** through an E-UTRAN **900b** and an SGW **900f**. In operation **903**, an MME **900d** may transmit a handover command to the E-UTRAN **900b**. In operation **905**, the E-UTRAN **900b** may transmit the handover command from the E-UTRAN **900b** to the UE **900a**. In operation **907**, the UE **900a** may transmit a confirmation of a handover to a 5G-RAN, for example, the NG-RAN **900c**, to the NG-RAN **900c**. The UE **900a** may move from the E-UTRAN **900b** and may perform synchronization with the target NG-RAN **900c**. In operation **909**, the UE **900a** may be in a state of being able to transmit an uplink user plane PDU (UL UP PDU) via the 5GS. DL UP PDUs and UL UP PDUs may be transmitted and received between the E-UTRAN **900b**, the SGW **900f**, and the UPF+PGW-U **900j** in operation **911** and operation **913**, and PDUs may be transmitted through the NG-RAN **900c** in operation **915**, for example, UP PDUs may be forwarded through a tunnel for PDU forwarding. The UE **900a** may resume uplink transmission of user plane data only for QFIs and session IDs having radio resources allocated to the NG-RAN **900c**. A v-UPF **900h** may forward PDUs to the NG-RAN **900c** using N3 tunnel information.

(110) According to various embodiments, in operation **917**, the NG-RAN **900c** may notify a target AMF **900e** that the UE **900a** is handed over to the NG-RAN **900c**. A notification message may include N2 SM information (e.g., N3 DL AN Tunnel Info). The target AMF **900e** may identify that the UE **900a** has arrived at a target side. In operation **919**, the target AMF **900e** may transmit a forward relocation complete notification message to the MME **900d**, thereby notifying that the UE **900a** has arrived at the target side. In operation **921**, the MME **900d** may transmit a forward relocation complete notification acknowledgement message to the target AMF (**900e**). In operation **923**, the target AMF **900e** may transmit a Nsmf_PDUSession_UpdateSMContext Request to the SMF+PGW-C **900i**. The Nsmf_PDUSession_UpdateSMContext Request may be a handover completion confirmation with respect to a PDU session ID. The handover completion may be transmitted to identify success of the N2 handover for each PDU session. In operation **925**, the SMF+PGW-C **900i** may perform N4 session modification with the SMF+PGW-U **900j**. The SMF+PGW-C **900i** may update the SMF+PGW-U **900j** on V-CN tunnel information, which indicates that a downlink user plane of a specified PDU session is changed to the NG-RAN **900c** and that a CN tunnel for an EPS bearer corresponding to the PDU session can be released. In operation **927**, the SMF+PGW-C **900i** may perform an SM policy association modification procedure with an h-PCF **9001**. When a PCC infrastructure is used, the SMF+PGW-C **900i** may report, for example, that a RAT type and a UE location are changed. In operation **929**, the SMF+PGW-C **900i** may transmit a Nsmf_PDUSession_UpdateSMContext Response, which may include a PDU session ID. The SMF+PGW-C **900i** may confirm reception of the handover completion. In operation **931**, a v-SMF **900g** may provide N3 DL AN Tunnel Info and N9 UL CN Tunnel Info to a v-UPF **900h**. In operation **935**, the UE **900a** may transmit and receive UL and DL UP PDUs to and from the UPF+PGW-U **900j** via the NG-RAN **900c**. In operation **937**, the UE **900a** may perform an EPS-to-5GS mobility registration procedure. In operation **939**, the MME

900d may clean up a resource in the EPC.

(111) FIG. **10** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure.

(112) According to various embodiments, the electronic device **101** (e.g., at least one of the processor **120** of FIG. **1**, FIG. **2A**, or FIG. **2B**, the first communication processor **212** of FIG. **2A**, the second communication processor **214** of FIG. **2A**, or the integrated communication processor **260** of FIG. **2B**) may perform a handover procedure from an EPS to a 5GS as in FIG. **9**. For example, as illustrated in FIG. **8B**, it is assumed that the electronic device **101** establishes one PDN connection of EBI=5 in a state of being registered in the EPS. Referring to FIG. **10**, in operation **1001**, an electronic device **101** may transmit a registration request message including EBI=5 to a first network **700** corresponding to the 5GS. The first network **700** may establish a PDU session of PSI=5, based on the EBI (e.g., EBI=5) in the registration request message. In operation **1003**, the first network **700** may transmit a registration accept message including information of PSI=5 to the electronic device **101**. In operation **1005**, the electronic device **101** may transmit a registration complete message to the first network **700**. Accordingly, the PDU session of PSI=5 may be established between the electronic device **101** and the first network **700**.

(113) According to various embodiments, in operation **1007**, the electronic device **101** may identify at least one third PDU session that is not included in at least one second PDU session among at least one first PDU session. For example, before an EPS fallback, the electronic device **101** stores information related to a PDU session of PSI=5 and a PDU session of PSI=6 in operation **795** of FIG. **7C**. The electronic device **101** may identify that the PDU session of PSI=6 does not exist based on establishment of the PDU session of PSI=5 after returning to the 5GS. In operation **1009**, the electronic device **101** may transmit a PDU session establishment request message for establishing the PDU session of PSI=6 to the first network **700**. The first network **700** may transmit a PDU session establishment accept message for the PDU session of PSI=6 to the electronic device **101** in operation **1011**. The PDU session establishment accept message may include information of PSI=6 and information of mapped EBI=6. The PDU session of PSI=5 and the PDU session of PSI=6 may be established between the electronic device **101** and the first network **700**.

Accordingly, the electronic device **101** may use the same PDU sessions before the EPS fallback and after returning to the 5GS and may thus continuously use a service.

(114) FIG. **11** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure.

(115) According to various embodiments, the electronic device **101** (e.g., at least one of the processor **120** of FIG. **1**, FIG. **2A**, or FIG. **2B**, the first communication processor **212** of FIG. **2A**, the second communication processor **214** of FIG. **2A**, or the integrated communication processor **260** of FIG. **2B**) may perform a handover procedure from an EPS to a 5GS as in FIG. **9**. For example, it is assumed that the electronic device **101** establishes a PDN connection of EBI=5 and a PDN connection of EBI=6 in a state of being registered in the EPS. The PDN connection of EBI=5 may correspond to a PDU session of PSI=5 before an EPS fallback, and the PDN connection of EBI=6 may correspond to a PDU session of PSI=6 before the EPS fallback. Referring to FIG. **11**, in operation **1101**, an electronic device **101** may transmit a registration request message including EBI=5 and EBI=6 to a first network **700** corresponding to the 5GS. For example, when a tracking area code (TAC) and/or a cell identifier are the same as before, the electronic device **101** may transmit a registration request message including an EBI used in the EPS. The first network **700** may establish a PDU session of PSI=5 and a PDU session of PSI=6, based on the EBIs (e.g., EBI=5 and 6) in the registration request message. In operation **1103**, the first network **700** may transmit a registration accept message including information of PSI=5 and 6 to the electronic device **101**. In operation **1105**, the electronic device **101** may transmit a registration complete message to the first network **700**. Accordingly, the PDU session of PSI=5 and the PDU session of PSI=6 may be established between the electronic device **101** and the first network **700**.

(116) According to various embodiments, in operation **1107**, the electronic device **101** may identify at least one fourth PDU session having the same identification information as at least one second PDU session among at least one first PDU session and having a different characteristic therefrom. For example, the electronic device **101** may identify that the QCI of the PDU session of PSI=6 before the EPS fallback, which is 9 does not match the QCI of the PDU session of PSI=6 after returning to the 5GS, which is 2. The electronic device **101** may determine to modify the characteristic of the PDU session, for example, the QCI, to be the same as that before the EPS fallback. In operation **1109**, the electronic device **101** may transmit a PDU session modification request message for modifying the QCI of the PDU session of PSI=6 to 9 to the first network **700**. The PDU session modification request message may include information of PSI=6 and information of QCI=9. The first network **700** may modify the QCI of the PDU session of PSI=6 to 9, for example, the QCI of a QoS flow in the PDU session to 9, based on the PDU session modification request message. In operation **1111**, the first network **700** may transmit a PDU session modification accept message to the electronic device **101**. The PDU session modification accept message may include the information of PSI=6 and the information of QCI=9. Accordingly, the electronic device **101** may use the PDU sessions of the same characteristic before the EPS fallback and after returning to the 5GS and may thus continuously use a service.

(117) FIG. **12** is a flowchart illustrating an operating method of an electronic device according to an embodiment of the disclosure.

(118) According to various embodiments, the electronic device **101** (e.g., at least one of the processor **120** of FIG. **1**, the first communication processor **212** of FIG. **2A**, the second communication processor **214** of FIG. **2A**, or the integrated communication processor **260** of FIG. **2B**) may perform a handover procedure from an EPS to a 5GS. Referring to FIG. **12**, when a first network **700** does not support an N26 interface, an electronic device **101** may transmit a registration request message not including an EBI to the first network **700** in operation **1201**. For example, whether the N26 interface is supported may be identified based on a flag of an information element (IE) of a 5GS network feature support. In operation **1203**, the electronic device **101** may receive a registration accept message from the first network **700**. In operation **1205**, the electronic device **101** may transmit a registration complete message to the first network **700**. When the N26 interface is supported, the electronic device **101** may transmit a PDU session establishment request message to the first network **700** in operation **1207** after registering with the 5GS. For example, the electronic device **101** may transmit a message for establishing a PDU session corresponding to a PDN connection, which is established in a state of being registered in the EPS, to the first network **700**. For example, the PDU session establishment request message may have a configuration of a handover type. In operation **1209**, the electronic device **101** may receive a PDU session establishment accept message from the first network **700**.

(119) According to various embodiments, in operation **1211**, the electronic device **101** may compare at least one first PDU session established before an EPS fallback and at least one PDU session established after returning to the 5GS. In operation **1213**, the electronic device **101** may perform an operation corresponding to a comparison result. When a PDU session not included in at least one second PDU session established after returning to the 5GS among the at least one first PDU session established before the EPS fallback is identified, the electronic device **101** may transmit a PDU session establishment request message for establishing the identified session. When a PDU session having the same identification information as the at least one second PDU session established after returning to the 5GS and having a different characteristic therefrom among the at least one first PDU session established before the EPS fallback, is identified, the electronic device **101** may transmit a PDU session modification request message for modifying the characteristic of the identified PDU session.

(120) According to various embodiments, an electronic device (e.g., the electronic device **101**) may include at least one processor (e.g., at least one of the processor **120**, the first communication

processor **212**, the second communication processor **214**, or the integrated communication processor **260**), wherein the at least one processor may be configured to establish at least one first PDU session in a state of being registered in a 5GS, store first information related to the at least one first PDU session, perform a procedure for registering with an EPS, based on at least one first trigger causing a system fallback from the 5GS to the EPS, perform a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, compare second information related to at least one second PDU session, established in association with registration with the 5GS, with the first information, perform an operation of establishing, based on a result of comparing the first information and the second information, at least one third PDU session when the at least one third PDU session not included in the at least one second PDU session among the at least one first PDU session is identified, and perform an operation of modifying, based on the result of comparing the first information and the second information, at least one fourth PDU session when the at least one fourth PDU session having a same identification information as the at least one second PDU session and having a characteristic different from a characteristic of the at least one second PDU session among the at least one first PDU session is identified.

(121) According to various embodiments, the at least one processor may be, as at least part of the performing of the procedure for registering with the EPS, configured to perform a tracking area update (TAU) procedure or an attach procedure with a network corresponding to the EPS.

(122) According to various embodiments, the at least one processor may be, as at least part the performing of the TAU procedure or the attach procedure, configured to transmit a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receive a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to portion of the at least one first PDU session, wherein the electronic device, in a state of being registered in the EPS, may establish the at least one PDN connection with the network corresponding to the EPS and may not establish a PDN connection corresponding to a remaining PDU session other than the portion of the at least one first PDU session.

(123) According to various embodiments, the at least one processor may be, as at least part of the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS, configured to transmit a registration request message including identification information about the at least one PDU session to a network corresponding to the 5GS, and receive a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, and wherein the registration accept message may include identification information corresponding to the at least one second PDU session.

(124) According to various embodiments, the at least one third PDU session may correspond to the remaining PDU session other than the portion of the at least one first PDU session, and wherein the at least one processor may be, as at least part of the establishing of the at least one third PDU session, configured to transmit a PDU session establishment request message including information stored in association with the remaining PDU session other than the portion of the at least one first PDU session to the network corresponding to the 5GS.

(125) According to various embodiments, the at least one processor may be, as at least part of the performing of the TAU procedure or the attach procedure, configured to transmit a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receive a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to the at least one first PDU session, wherein the electronic device, in a state of being registered in the EPS, may establish the at least one PDN connection

with the network corresponding to the EPS.

(126) According to various embodiments, the at least one processor may be, as at least part of the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS, configured to transmit a registration request message including identification information about the at least one PDU session to a network corresponding to the 5GS, and receive a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, wherein the registration accept message may include identification information corresponding to the at least one second PDU session, wherein the at least one fourth PDU session among the at least one first PDU session may have the same identification information as at least portion of the at least one second PDU session and may have a different characteristic therefrom.

(127) According to various embodiments, the at least one processor may be, as at least part of the modifying of the at least one fourth PDU session, configured to transmit, to the network corresponding to the 5GS, a PDU session modification request message including identification information corresponding to the at least one fourth PDU session and characteristic of the at least one fourth PDU session.

(128) According to various embodiments, the at least one processor may be, as at least part the performing of the procedure for registering with the 5GS, configured to transmit, to a network corresponding to the 5GS when the network corresponding to the 5GS supports an N26 interface, a registration request message including identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS.

(129) According to various embodiments, the at least one processor may be further configured to transmit, to a network corresponding to the 5GS, based on the network corresponding to the 5GS does not supporting an N26 interface, a PDU session establishment request message including identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS, in the state of being registered in the 5GS.

(130) According to various embodiments, an operating method of an electronic device (e.g., the electronic device **101**) may include establishing at least one first PDU session in a state of being registered in a 5GS, storing first information related to the at least one first PDU session, performing a procedure for registering with an EPS, based on at least one first trigger causing a system fallback from the 5GS to the EPS, performing a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, comparing second information related to at least one second PDU session, established in association with registration with the 5GS, with the first information, performing an operation of establishing, based on a result of comparing the first information with the second information, at least one third PDU session when the at least one third PDU session not included in the at least one second PDU session among the at least one first PDU session is identified, and performing an operation of modifying, based on the result of comparing the first information with the second information, at least one fourth PDU session when the at least one fourth PDU session having a same identification information as the at least one second PDU session and having a characteristic different from a characteristic of the at least one second PDU session among the at least one first PDU session is identified.

(131) According to various embodiments, the performing of the procedure for registering with the EPS performs a tracking area update (TAU) procedure or an attach procedure with a network corresponding to the EPS.

(132) According to various embodiments, the performing of the TAU procedure or the attach procedure may include transmitting a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receiving a TAU accept message according

to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to portion of the at least one first PDU session, and wherein the electronic device, in a state of being registered in the EPS, may establish the at least one PDN connection with the network corresponding to the EPS and may not establish a PDN connection corresponding to a remaining PDU session other than the portion of the at least one first PDU session.

(133) According to various embodiments, the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS may include transmitting a registration request message including identification information about the at least one PDU session to a network corresponding to the 5GS, and receiving a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, wherein the registration accept message may include identification information corresponding to the at least one second PDU session.

(134) According to various embodiments, the at least one third PDU session may correspond to the remaining PDU session other than the portion of the at least one first PDU session, wherein the performing of the operation of establishing the at least one third PDU session may include transmitting a PDU session establishment request message including information stored in association with the remaining PDU session other than the portion of the at least one first PDU session to the network corresponding to the 5GS.

(135) According to various embodiments, the performing of the TAU procedure or the attach procedure may include transmitting a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receiving a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to the at least one first PDU session, wherein the electronic device, in a state of being registered in the EPS, may establish the at least one PDN connection with the network corresponding to the EPS.

(136) According to various embodiments, the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS may include transmitting a registration request message including identification information about the at least one PDU session to a network corresponding to the 5GS, and receiving a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, wherein the registration accept message may include identification information corresponding to the at least one second PDU session, wherein the at least one fourth PDU session among the at least one first PDU session may have the same identification information as at least portion of the at least one second PDU session and may have a different characteristic therefrom.

(137) According to various embodiments, the performing of the operation of modifying the at least one fourth PDU session transmits, to the network corresponding to the 5GS, a PDU session modification request message including the identification information corresponding to the at least one fourth PDU session and the characteristic of the at least one fourth PDU session.

(138) According to various embodiments, the performing of the procedure for registering with the 5GS transmits, to a network corresponding to the 5GS, based on the network corresponding to the 5GS supporting an N26 interface, a registration request message including identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS.

(139) According to various embodiments, the method may further include transmitting, to a network corresponding to the 5GS, based on the network corresponding to the 5GS does not supporting an N26 interface, a PDU session establishment request message including identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS in the state of being

registered in the 5GS.

(140) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(141) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(142) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(143) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(144) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(145) According to various embodiments, each component (e.g., a module or a program) of the

above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(146) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: memory storing one or more computer programs; and one or more processors communicatively coupled to the memory, wherein the one or more computer programs include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: establish at least one first protocol data unit (PDU) session in a state of being registered in a 5.sup.th generation system (5GS), store first information related to the at least one first PDU session, wherein the first information includes at least one of network information of the at least one first PDU session, first PDU session information including first PDU session ID (PSI) of the at least one first PDU session, or first QoS information of at least one first PDU session, perform a procedure for registering with an evolved packet system (EPS), based on at least one first trigger causing a system fallback from the 5GS to the EPS, perform a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS, compare second information, related to a second PDU session, established in association with registration with the 5GS, with the first information, wherein the second information includes at least one of network information on the second PDU session, second PDU session information including second PSI of the second PDU session, or second QoS information of the second PDU session, when the first PSI is different from the second PSI, establish at least one third PDU session based on the first information, and when 1) the first PSI corresponds to the second PSI and 2) a characteristic of the at least one first PDU session and a characteristic of the second PDU session are different from each other, modify the second PDU session based on the first information, wherein the characteristic of the at least one first PDU session includes at least one of single network slice selection assistance information (S-NSSAI) of the at least one first PDU session, QoS class identifier (QCI) of the at least one first PDU session, or data network name (DNN) of the at least one first PDU session, and wherein the characteristic of the second PDU session includes at least one of S-NSSAI of the second PDU session, QCI of the second PDU session, or data network name (DNN) of the second PDU session.

2. The electronic device of claim 1, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to perform a tracking area update (TAU) procedure or an attach procedure with a network corresponding to the EPS.

3. The electronic device of claim 2, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit a TAU request message according to the TAU procedure or an attach

request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receive a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one packet data network (PDN) connection corresponding to a portion of the at least one first PDU session, and wherein the electronic device, in a state of being registered in the EPS, establishes the at least one PDN connection with the network corresponding to the EPS and does not establish a PDN connection corresponding to a remaining PDU session other than the portion of the at least one first PDU session.

4. The electronic device of claim 3, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit a registration request message comprising identification information about the at least one first PDU session to a network corresponding to the 5GS, and receive a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, and wherein the registration accept message includes identification information corresponding to the second PDU session.

5. The electronic device of claim 4, wherein the at least one third PDU session corresponds to the remaining PDU session other than the portion of the at least one first PDU session, and wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to transmit a PDU session establishment request message comprising information stored in association with the remaining PDU session other than the portion of the at least one first PDU session to the network corresponding to the 5GS.

6. The electronic device of claim 2, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receive a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to the at least one first PDU session, and wherein the electronic device, in a state of being registered in the EPS, establishes the at least one PDN connection with the network corresponding to the EPS.

7. The electronic device of claim 6, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit a registration request message comprising identification information about the at least one PDU session to a network corresponding to the 5GS, and receive a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, wherein the registration accept message includes identification information corresponding to the at least one second PDU session, and wherein the modified second PDU session has the same identification information as at least portion of the second PDU session and has a characteristic different from the characteristic of the second PDU session among the at least one first PDU session.

8. The electronic device of claim 7, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to transmit, to the network corresponding to the 5GS, a PDU session modification request message comprising identification information corresponding to the modified second PDU session and characteristic of the modified second PDU session.

9. The electronic device of claim 1, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit, to a network corresponding to the 5GS, based on the network corresponding to the 5GS supporting an N26 interface, a registration request message comprising

identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS, in a state of being registered in the EPS.

10. The electronic device of claim 1, wherein the one or more computer programs further include computer-executable instructions that, when executed by the one or more processors, cause the electronic device to: transmit, to a network corresponding to the 5GS, based on the network corresponding to the 5GS does not supporting an N26 interface, a PDU session establishment request message comprising identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS, in a state of being registered in the EPS, in the state of being registered in the 5GS.

11. A non-transitory storage medium storing one or more program, the one or more program comprising computer-executable instructions, when executed by at least one processor an electronic device, cause the electronic device to: establish at least one first protocol data unit (PDU) session in a state of being registered in a 5.sup.th generation system (5GS); store first information related to the at least one first PDU session, wherein the first information includes at least one of network information of the at least one first PDU session, first PDU session information including first PDU session ID (PSI) of the at least one first PDU session, or first QoS information of at least one first PDU session; perform a procedure for registering with an evolved packet system (EPS), based on at least one first trigger causing a system fallback from the 5GS to the EPS; perform a procedure for registering with the 5GS, based on at least one second trigger causing a return to the 5GS; compare second information related to a second PDU session, established in association with registration with the 5GS, with the first information, wherein the second information includes at least one of network information on the second PDU session, second PDU session information including second PSI of the second PDU session, or second QoS information of the second PDU session; when the first PSI is different from the second PSI, establish at least one third PDU session based on the first information; and when 1) the first PSI corresponds to the second PSI and 2) a characteristic of the at least one first PDU session and a characteristic of the second PDU session are different from each other, modify the second PDU session based on the first information, wherein the characteristic of the at least one first PDU session includes at least one of S-NSSAI of the at least one first PDU session, QCI of the at least one first PDU session, or data network name (DNN) of the at least one first PDU session, and wherein the characteristic of the second PDU session includes at least one of S-NSSAI of the second PDU session, QCI of the second PDU session, or data network name (DNN) of the second PDU session.

12. The non-transitory storage medium of claim 11, wherein the performing of the procedure for registering with the EPS performs a tracking area update (TAU) procedure or an attach procedure with a network corresponding to the EPS.

13. The non-transitory storage medium of claim 12, wherein the performing of the TAU procedure or the attach procedure comprises: transmitting a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receiving a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to a portion of the at least one first PDU session, and wherein the electronic device, in a state of being registered in the EPS, establishes the at least one PDN connection with the network corresponding to the EPS and does not establish a PDN connection corresponding to a remaining PDU session other than the portion of the at least one first PDU session.

14. The non-transitory storage medium of claim 13, wherein the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS comprises: transmitting a registration request message comprising identification information about the at least one first PDU session to a network corresponding to the 5GS, and receiving a registration accept message corresponding to the registration request message from the network

corresponding to the 5GS, and wherein the registration accept message includes identification information corresponding to the at least one second PDU session.

15. The non-transitory storage medium of claim 14, wherein the at least one third PDU session corresponds to the remaining PDU session other than the portion of the at least one first PDU session, and wherein the performing of the operation of establishing the at least one third PDU session includes transmitting a PDU session establishment request message comprising information stored in association with the remaining PDU session other than the portion of the at least one first PDU session to the network corresponding to the 5GS.

16. The non-transitory storage medium of claim 12, wherein the performing of the TAU procedure or the attach procedure comprises: transmitting a TAU request message according to the TAU procedure or an attach request message according to the attach procedure, which includes identification information corresponding to the at least one first PDU session, and receiving a TAU accept message according to the TAU procedure or an attach accept message according to the attach procedure, which includes identification information about at least one PDN connection corresponding to the at least one first PDU session, and wherein the electronic device, in a state of being registered in the EPS, establishes the at least one PDN connection with the network corresponding to the EPS.

17. The non-transitory storage medium of claim 16, wherein the performing of the procedure for registering with the 5GS, based on the at least one second trigger causing the return to the 5GS comprises: transmitting a registration request message comprising identification information about the at least one PDU session to a network corresponding to the 5GS, and receiving a registration accept message corresponding to the registration request message from the network corresponding to the 5GS, wherein the registration accept message includes identification information corresponding to the at least one second PDU session, and wherein the modified second PDU session has the same identification information as at least portion of the at least one second PDU session and has a characteristic different from the characteristic of the at least one second PDU session among the at least one first PDU session.

18. The non-transitory storage medium of claim 17, wherein the performing of the operation of modifying the modified second PDU session transmits, to the network corresponding to the 5GS, a PDU session modification request message comprising the identification information corresponding to the modified second PDU session and the characteristic of the modified second PDU session.

19. The non-transitory storage medium of claim 11, wherein the performing of the procedure for registering with the 5GS transmits, to a network corresponding to the 5GS, based on the network corresponding to the 5GS supporting an N26 interface, a registration request message comprising identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS.

20. The non-transitory storage medium of claim 11, further comprising: transmitting, to a network corresponding to the 5GS, based on the network corresponding to the 5GS does not supporting an N26 interface, a PDU session establishment request message comprising identification information about at least one PDN connection established between the electronic device and a network corresponding to the EPS in a state of being registered in the EPS, in the state of being registered in the 5GS.
